

M13 / ASSESSMENT

M13 Module Revision Test

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Read this first

M13 | DELTA LAKE + LAKEHOUSE

CLOSED-BOOK MODULE REVISION TEST

Attempt after the learner book, cluster checks and practice. Keep the answer book closed. Untimed mastery first.

Course runtime baseline: Spark 4.2.0 + Delta Lake 4.4.0. Live Delta/PySpark execution remains learner-PC proof unless explicitly recorded.

Instructions

Use your own words. Where a question asks for proof, include exact controls. For code questions, pseudocode is acceptable only when the question says so; otherwise write executable-style PySpark/Delta logic.

1. [Foundations] Explain the difference between a data lake, warehouse and lakehouse without saying one is always better.
2. [Files vs tables] Why can a raw scan of every Parquet file in a Delta table directory return the wrong logical state?
3. [Transaction state] A job creates files and crashes. What evidence tells you whether a new table state committed?
4. [ACID] Give one concrete failure/visibility example for atomicity and isolation.
5. [DML] Design a safe DELETE workflow for a known test population.
6. [Schema] When should a new column be rejected, quarantined or evolved?
7. [Time travel] Differentiate historical read, RESTORE and VACUUM.
8. [MERGE] What conditions are necessary before you can call a MERGE-based load idempotent?
9. [Ambiguity] Two source rows share one key and identical latest timestamps but disagree on amount. What should happen?
10. [SCD2] State three invariants for a correct SCD Type 2 dimension.
11. [Medallion] Give two reconciliation equations across Bronze, Silver, quarantine and Gold.
12. [Maintenance] Why does OPTIMIZE not fix duplicate business rows?
13. [Maintenance] Why can aggressive VACUUM damage recovery/time-travel capability?
14. [Incident] Gold total is wrong but Bronze count matches source. Give an evidence-first debugging order.
15. [Runtime honesty] What is the difference between static/simulator QA and learner-PC Delta runtime proof?

Practical section

- P1. Starting from a target orders table and a source batch, write MERGE logic that updates only newer records and inserts new keys. Add a duplicate-source-key precheck.
- P2. Write a reconciliation plan for an SCD2 load with changed/new/unchanged customers.
- P3. Given Bronze 120 rows, Silver valid 115, quarantine 5 and Gold revenue 91,250, state the minimum controls needed before calling the run successful.
- P4. A table has many small files and an accidental bad delete from yesterday. Decide whether to OPTIMIZE, VACUUM, time travel or restore first, and justify the order.
- P5. Draw an end-to-end run contract from source batch -> Bronze -> Silver -> curated MERGE -> Gold -> checkpoint, showing when external progress may be committed.